Omer El Idrissi

Systems Programmer, Linux Contributor

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github.com/omeridrissi/

Projects

fifine_mic_driver
Linux microphone driver. It communicates with the microphone's USB isochronous endpoint and efficiently delivers the pure uncompressed stream with a fast single-consumer, multiple producers ring buffer model.

prongc
It's a static C cross-function variable dependency analysis tool that checks if two or more functions touch the same data. It can check if the shared data has the same lock protections. It supports multithreaded parsing and compile_command.json. Depends on libclang for AST analysis.

edu_dev
QEMU edu virtual device PCI driver. IRQ handling, DMA, MMIO.

drm_demo
libdrm framebuffer demonstration.

cachepix
Cachepix is a high-performance C library for processing PPM (portable pixmap) images, with a strong focus on data layout, cache efficiency, and SIMD optimization.

Skills

C Programming

Linux

Core Competencies
- Systems programming
- Device driver analysis

Open Source Contributions

Linux kernel
- [staging]: Cleaned up vendor-defined error macros, replacing with proper kernel errno codes (2-patch series)
- [staging]: Fixed style mistake and minor type fix

Education

Self-directed study for systems programming & kernel development

Kernel Development:
- Learned kernel module architecture, USB, PCI, DRM GEM basics, KMS through linux kernel documentation.

Performance & Architecture:
- Studied memory hierarchy, caching, and code optimization from Computer Systems: A Programmer's Perspective (Bryant & O'Hallaron).

Hardware & Electronics Basics:
- Worked through Practical Electronics for Inventors (Scherz & Monk), developed ability to read and interpret chip datasheets and USB device specifications.